

Role Based Dashboard:

Store Manager:

The Store Manager is responsible for the day-to-day operations of a specific store.

Their focus is:

- Is the store running smoothly?
- Are customers interacting with products?
- Which shelves need attention?
- Are cameras working?
- Is there overcrowding?

They **do not need** deep AI analytics or campaign performance.

Section 1 – KPI Cards:

- Today's Visitors
- Current Customers in Store
- Average Dwell Time
- Products Picked Today
- Purchase Conversion Rate
- Online Cameras

Section 2 – Live Store Cameras: Only camera assigned to their stores.

Each camera should display:

- Live Video
- People Count
- Current Zone
- Crowd Status
- Shelf Activity
- Camera Health

Example:

- **Entrance camera**
- **LIVE**
- **People: 18**
- **Crowd: Medium**
- **Status: Online**

Section 3 – Store Traffic:

Charts

- Visitors by Hour (Line Chart)
- Customers by Zone (Bar Chart)

Purpose

Shows where customers are currently located.

Section 4 – Shelf Performance:

Charts

- Most Visited Shelf
- Least Visited Shelf
- Shelf Engagement Score

Example

Shelf A
92%

Shelf B
74%

Shelf C
38%

Section 5 – Product Interaction:

Displays

- Most Picked Products
- Most Returned Products
- Least Viewed Products
- Most Compared Products

Section 6 – Store Alerts:

- Camera Offline
- Shelf Receiving Low Attention
- Overcrowded Area
- Long Checkout Queue
- Product Out of Stock

Section 7 – Store Heatmap:

- Customer Traffic
- Busy Areas
- Empty Areas
- Store Manager only sees their own store.

2. Retail Analyst Dashboard

Role Objective: Retail Analysts analyze **customer behavior and shopping patterns**.

Section 1 – KPI Cards

- Average Attention Time
- Average Dwell Time
- Repeat Visitors
- Average Session Length
- Customer Segments
- Engagement Score

Section 2 – Consumer Journey Analysis:

Charts:

Customer Flow

Entry

↓

Aisle 1

↓

Shelf B

↓

Checkout

Purpose: Understand customer movement.

Section 3 – Attention Analytics: Charts

- Average Attention per Shelf
- Average Attention per Product
- Attention by Time

Section 4 – Customer Segmentation:

- Pie Chart
- Explorers
- Quick Buyers
- Comparison Buyers
- Impulse Buyer
- Brand Loyal Customers

Section 5 – Heatmaps:

- Displays
- Store Heatmap
- Shelf Heatmap
- Traffic Heatmap
- Attention Heatmap

Section 6 – Shopping Behavior: Charts

- Most Viewed Products
- Most Compared Products
- Longest Viewed Products
- Frequently Ignored Products

Section 7 – Customer Journey:

Sankey Diagram

Entrance



Beverages



Snacks



Billing

A **Sankey diagram** is a type of data visualization used to show the **flow of items, energy, data, or people** from one stage or category to another. Instead of showing static totals, it illustrates **movement and process paths**.



Depending on the tech stack assigned , You will implement the diagram using one of the following methods:

- **No-Code / Rapid Mockup:** Input formatted text directly into **SankeyMATIC** for quick prototype generation.
- **BI Dashboard Stack:** If building in **Power BI** or **Tableau**, add the *Sankey Chart* visual component and bind the **Source**, **Target**, and **Weight/Value** fields to the dataset.
- **Code-Based Frontend:** If using **Python/Plotly** or **JavaScript (D3.js / Google Charts)**, pass the node indices and flow arrays directly into the visualization layout component.

Retail Analyst does not see

- User Management
- Camera Management
- Marketing Campaigns
- System Monitoring

3. Marketing Manager Dashboard: Marketing Managers evaluate campaign effectiveness and product visibility.

Section 1 – KPI Cards

Campaign Reach

Promotion Engagement

Product Visibility

Conversion Rate

Average Product Attractiveness Score

Campaign ROI

Section 2 – Campaign Performance: Charts

Campaign A

Campaign B

Campaign C {Use: **Bar Chart**}

Measures:

- Customer Engagement
- Sales Lift
- Attention Generated

Section 3 – Product Visibility: Charts

- Visibility Score

- Attention Score
- Pickup Rate
- Purchase Rate

Section 4 – Promotion Performance: Displays

Before Promotion

↓

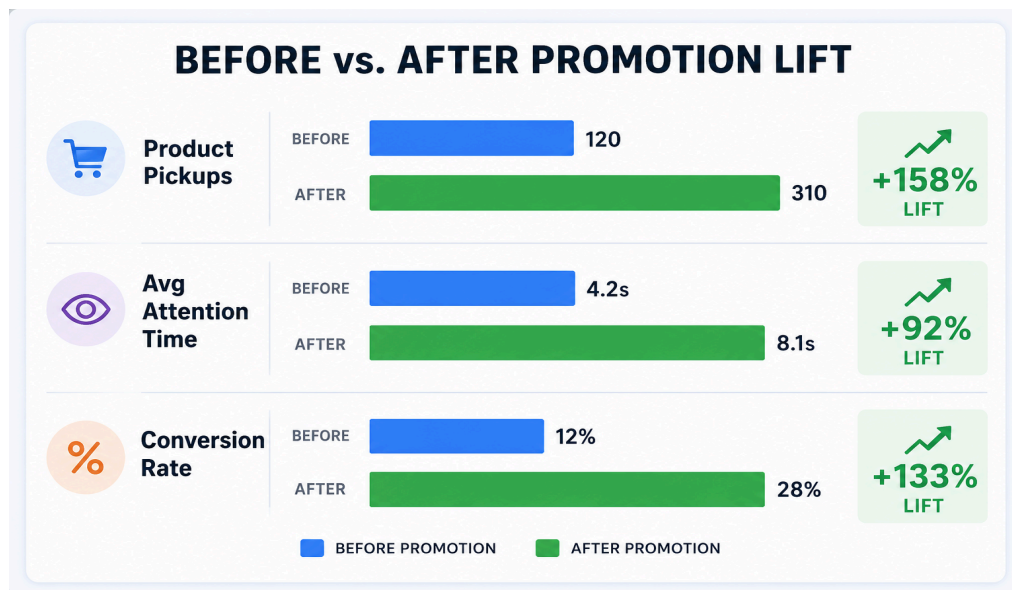
After Promotion

Comparison Charts

Dual-Bar / Grouped Bar Chart (Best for Metrics Comparison):

Display key metrics side-by-side for the baseline period vs. the promotional period.

- **Categories on X-axis:** Attention Time (s), Product Pickups, Checkout Conversions.
- **Bars per category:**
 - **Blue Bar:** Before Promotion
 - **Green Bar:** After Promotion (with a percentage badge showing the % Lift).



Section 5 – Product Attractiveness: Displays

- Top Attractive Products
- Least Attractive Products
- Product Ranking

Section 6 – Shelf Comparison:

Charts:

- Shelf A
- Shelf B
- Shelf C

Shows:

- Attention
- Conversion
- Engagement

Section 7 – AI Recommendations: Displays

- Move Product A to Shelf 2
- Increase Shelf Lighting
- Change Promotional Banner
- Move Display Near Entrance
- Replace Underperforming Promotion

Marketing Manager **does not see**

- Live Camera Monitoring
- User Management
- System Health
- API Logs

4. Administrator Dashboard:

The Administrator manages **the platform**, not the retail business.

They are responsible for:

- User accounts
- Camera infrastructure
- System performance
- Security
- Platform availability

Section 1 – KPI Cards:

- Total Users
- Total Stores
- Total Cameras

- Running Services
- System Uptime
- API Requests

Section 2 – Management

- Table
- User
- Role
- Store
- Status
- Last Login
- Actions

Section 3 – Camera Management: Displays

- Camera List
- Store Assignment
- Resolution
- Status
- FPS
- Restart Camera
- Assign Shelf

Section 4 – Platform Monitoring

Charts:

- CPU Usage
- GPU Usage
- RAM Usage
- Database Load
- API Response Time
- Storage Usage

Section 5 – Camera Health

Displays

- Online Cameras
- Offline Cameras
- Recording Status

- Streaming Status
- Network Quality

Section 6 – Security Logs: Displays

Login Attempts

Failed Logins

Permission Changes

Password Updates

System Errors

Section 7 – Audit Logs: Displays

- User Created
- Camera Added
- Store Update
- Shelf Modified
- Role Changed

Administrator **does not focus on**

- Customer behavior analysis
- Product attractiveness
- Marketing campaign performance
- Consumer journey analytics

Charts You have to implemented in Your project:

1. Store Manager Dashboard:

Section 1 – Store Traffic

- Line Chart (Hourly Visitor Trend)
- Area Chart (Daily Store Footfall Trend)

Section 2 – Zone Occupancy

- Vertical Bar Chart (Visitors per Zone)
- Donut Chart (Zone Occupancy Distribution)

Section 3 – Shelf Performance

- Horizontal Bar Chart (Shelf Engagement Score)
- Stacked Bar Chart (Viewed vs Picked vs Purchased)
- Heatmap (Shelf Attention Heatmap)

Section 4 – Product Interaction

- Horizontal Bar Chart (Top Picked Products)
- Horizontal Bar Chart (Most Returned Products)
- Horizontal Bar Chart (Most Compared Products)
- Line Chart (Product Pickup Trend)

Section 5 – Store Conversion

- Funnel Chart (Entry → View → Pickup → Purchase)
- Gauge Chart (Store Conversion Rate)

Section 6 – Camera Monitoring

- Camera Grid View (Live Feed)
- Donut Chart (Camera Health Status)

Section 7 – Alerts

- Timeline Chart (Recent Alerts)
- Alert Cards (No Chart)

2. Retail Analyst Dashboard

Section 1 – Consumer Attention Analytics

- Line Chart (Average Attention Duration)
- Area Chart (Attention Trend)
- Box Plot (Attention Time Distribution)

Section 2 – Consumer Journey

- Sankey Diagram (Customer Journey)
- Flow Diagram (Movement Flow)

Section 3 – Customer Segmentation

- Pie Chart (Customer Segments)
- Donut Chart (Segment Distribution)

Section 4 – Shopping Behaviour

- Horizontal Bar Chart (Most Viewed Products)
- Horizontal Bar Chart (Most Ignored Products)
- Horizontal Bar Chart (Most Compared Products)
- Tree Map (Product Category Interest)

Section 5 – Heatmaps

- Customer Traffic Heatmap
- Store Attention Heatmap
- Shelf Heatmap
- Zone Heatmap

Section 6 – Dwell Time Analysis

- Violin Plot (Dwell Time Distribution)
- Line Chart (Average Dwell Time by Hour)

Section 7 – Behavioral Analytics

- Scatter Plot (Attention vs Purchase)
- Bubble Chart (Attention vs Dwell Time vs Conversion)

3. Marketing Manager Dashboard

Section 1 – Campaign Performance

- Grouped Bar Chart (Campaign Comparison)
- Line Chart (Campaign Performance Trend)

Section 2 – Promotion Effectiveness

- Before vs After Comparison Bar Chart
- Waterfall Chart (Sales Lift)
- Funnel Chart (Campaign Conversion)

Section 3 – Product Visibility

- Radar Chart (Visibility Metrics)
- Horizontal Bar Chart (Visibility Score)
- Heatmap (Shelf Visibility)

Section 4 – Product Attractiveness

- Horizontal Bar Chart (Product Attractiveness Ranking)
- Radar Chart (Attractiveness Score Breakdown)

Section 5 – Customer Engagement

- Line Chart (Engagement Trend)
- Donut Chart (Engagement Distribution)

Section 6 – Conversion Analysis

- Scatter Plot (Attention vs Conversion)
- Bubble Chart (Engagement vs Sales)

Section 7 – Marketing Recommendations

- Decision Matrix
- Priority Matrix (High Impact vs Low Impact)

4. Administrator Dashboard

Section 1 – System Overview

- Line Chart (System Uptime)
- Area Chart (API Requests Over Time)

Section 2 – User Analytics

- Pie Chart (Users by Role)
- Bar Chart (Active Users per Store)

Section 3 – Camera Monitoring

- Donut Chart (Camera Status)
- Bar Chart (Cameras by Store)
- Camera Grid View (Live Feed)

Section 4 – Infrastructure Monitoring

- Line Chart (CPU Usage)
- Line Chart (Memory Usage)
- Line Chart (GPU Usage)
- Line Chart (Disk Usage)
- Line Chart (Network Traffic)

Section 5 – API Performance

- Line Chart (API Response Time)
- Bar Chart (Endpoint Response Time)
- Area Chart (Request Volume)

Section 6 – Security Monitoring

- Line Chart (Login Attempts)
- Bar Chart (Failed Logins)
- Donut Chart (Authentication Status)

Section 7 – Audit Logs

- Timeline Chart (System Events)
- Activity Feed (Audit History)

This is the layout commonly used in enterprise BI tools such as **Power BI**, **Tableau**, **Grafana**, and AI-powered retail analytics platforms, and it fits the modules defined in your project specification.

The **recommended chart types as the default design, not as mandatory requirements**. The goal is to choose the chart that best represents the data in each analytics section. If they find a more suitable visualization that clearly communicates the same information, they can justify using it.

Dashboard Design Guidelines:

- Use the recommended chart type for each analytics section.
- Do **not** convert KPI cards, live camera feeds, heatmaps, tables, alerts, or timelines into charts.
- Each dashboard should have a balanced layout with KPIs, charts, tables, and operational widgets.
- Ensure every chart directly supports the role's responsibilities and decision-making.
- Use responsive and interactive charts (hover tooltips, filtering, drill-down where appropriate).

